SOIL SENSOR

SMD BOM & FOOTPRINT REVISION - v1.2

Modernized assembly plan: 0805 passive components + SOD-123 diode + through-hole connector

1. Revision Decision

Locked: the design is being changed from the original all-THT passive-component strategy to an SMD-heavy assembly. Resistors and ceramic capacitors use **0805**. The 3-pin Arduino-facing connector J1 remains **through-hole** for repeated mechanical use. The 555 timer U1 remains **DIP-8 THT**.

Reasoning: 0805 is selected as the practical hand-soldering size for this project: larger and easier to inspect than smaller SMD packages, while remaining compact for a modern PCB. J1 stays THT because it will be repeatedly connected to jumper wires and benefits from through-hole mechanical strength.

2. Updated BOM / Footprint Lock

Ref.	Description	Value / Part	Package	Planned KiCad footprint
U1	555 Timer	NE555P	DIP-8 (THT)	Package_DIP:DIP-8_W7.62mm
R1	Resistor	1 kOhm	0805 SMD	Resistor_SMD:R_0805_2012Metric
R2	Resistor	10 kOhm	0805 SMD	Resistor_SMD:R_0805_2012Metric
R3	Resistor	10 kOhm	0805 SMD	Resistor_SMD:R_0805_2012Metric
R4	Resistor	100 kOhm	0805 SMD	Resistor_SMD:R_0805_2012Metric
C1	Ceramic capacitor	1 nF	0805 SMD	Capacitor_SMD:C_0805_2012Metric
C2	Ceramic capacitor	100 nF	0805 SMD	Capacitor_SMD:C_0805_2012Metric
C3	Ceramic capacitor	10 nF	0805 SMD	Capacitor_SMD:C_0805_2012Metric
C4	Capacitor	100 nF - 1 uF	0805 SMD	Capacitor_SMD:C_0805_2012Metric
D1	Small-signal diode	1N4148W (or equivalent)	SOD-123 SMD	Diode_SMD:D_SOD-123
J1	3-pin header	2.54 mm	THT	Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical

Footprint note: the footprint strings above are the planned KiCad library choices for the revision. Before PCB layout, verify the actual physical parts in hand against pad geometry and package drawing.

3. D1 Package Decision

D1 is moved to **SOD-123** as the default SMD package because it is relatively easy to hand-place and solder compared with smaller diode packages. The intended electrical part is **1N4148W or an equivalent small-signal diode**.

Important verification: the exact diode in hand must actually use the selected package (or be mechanically/electrically matched to it). Do not finalize the footprint from the part number alone. Confirm package drawing, pin numbering, polarity marking, and pad layout before PCB fabrication.

Diode polarity used by the schematic

```
D1 Pin 2 = Anode (A) -> NODE-X     |     D1 Pin 1 = Cathode (K) -> NODE-Y / AOUT
```

4. 0805 Passive Strategy

R1-R4 and C1-C4 are planned as 0805 SMD parts. In KiCad this corresponds to the 2012 metric package designation. The purpose is easier manual assembly and inspection while keeping the board substantially more compact than an all-through-hole implementation.

Group	Refs.	Package	KiCad footprint
Resistors	R1, R2, R3, R4	0805	Resistor_SMD:R_0805_2012Metric
Ceramic capacitors	C1, C2, C3	0805	Capacitor_SMD:C_0805_2012Metric
Output capacitor	C4	0805	Capacitor_SMD:C_0805_2012Metric

5. J1 Remains Through-Hole

J1 remains a 3-pin, 2.54 mm through-hole header. This is a deliberate mechanical decision because the connector is expected to be connected and disconnected repeatedly. The interface remains: **Pin 1 = VCC, Pin 2 = GND, Pin 3 = AOUT**.

The external ICD specifies VCC to Arduino 5V, GND to Arduino GND, and AOUT to Arduino A0.

6. Effect on Manufacturing

Moving the passive network to SMD changes the physical assembly strategy. The original design was selected for CNC/DIY fabrication and hand soldering; this revision keeps the mechanically stressed connector through-hole while reducing most passive-component drilling.

Expected drill usage: the SMD resistors, SMD capacitors, and SOD-123 diode do not need component through-holes. J1 still requires its three connector holes, and any future mounting holes will also require drilling. The copper sensing electrodes remain PCB copper features.

7. Schematic-to-PCB Handoff Checklist

1. Confirm all symbols are fully annotated (U1, R1-R4, C1-C5, D1, J1).
2. Set the planned footprints on every component before PCB layout.
3. Verify U1 footprint matches the actual NE555P package in hand.
4. Verify the exact D1 package and polarity marking before committing the SOD-123 footprint.
5. Verify 0805 resistor and capacitor bodies/pad geometry against the selected parts.
6. Keep J1 as a through-hole 2.54 mm connector and verify its mechanical orientation.
7. Keep C2 close to U1 during PCB placement.
8. Keep the sensing-electrode geometry as a PCB feature; do not treat it as a purchased component.
9. Re-run electrical checks and then perform PCB design-rule checking after footprint assignment and placement.

8. Project Electrical Interface - Reminder

The custom sensor board still exposes the same Arduino-facing interface: **VCC -> Arduino 5V, GND -> Arduino GND, AOUT -> Arduino A0**. The custom PCB sensing combs are the actual moisture-sensitive structure; the schematic capacitor symbol is only their electrical representation.

9. Revision Status

Revision	Status	Locked decisions
v1.2	Current	R1-R4 and C1-C4 = 0805 SMD; D1 = SOD-123 SMD default; U1 = DIP-8 THT; J1 = THT 2.54 mm header.

Source basis: project BOM/design notes and the current schematic/netlist, plus the explicit SMD migration decisions recorded for this revision.